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Follow the format and Instructions



Green University of Bangladesh
Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Fall, Year:2024), B.Sc. in CSE (Day)

Lab Report NO: 01
Course Title: Study of diode I-V characteristic
Course Code: EEE-202 Section:231 (D1)

Lab Experiment Name: Study of the diode I-V characteristic.

Student Details

	Name	ID
1.	Promod Chandra Das	231002005

Lab Date : 03-09-2024
Submission Date : 10-09-2024
Course Teacher's Name : Sakib Abdul Ahad

Lab Report Status

Marks:
Comments:.....

Signature:.....
Date:.....

Major Connections.

Inst 25
24.09.24

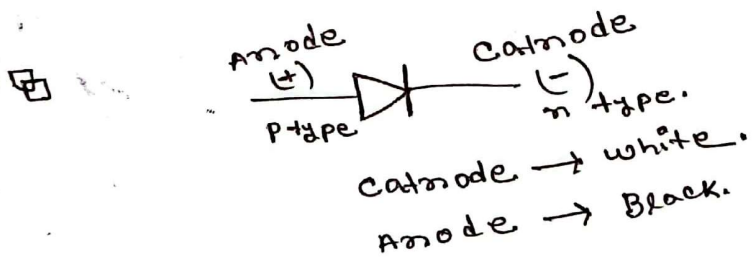
⇒ Experiment: Study of diode I-V characteristics

⇒ Objectives:—

To study the diode I-V characteristics.

⇒ Equipment Required:—

1. Resistor: $1.2\text{ K}\Omega$,
2. Diode,
3. Ammeter ($0-10\text{ A}$)
4. Multimeter.
5. Trainer Board.
6. Connecting wires.



power step - horizontally.
Other step - vertical.

Theory: Diode

- * A two terminal device that acts as a one way conductor.
 - * Built by connecting a p-type semiconductor with an n-type semiconductor
 - * The p-type side is called anode and the n-type side is called cathode.
 - * A diode has two connection configurations forward bias and Reverse Bias.
- ⇒ Forward bias connection.
 - ⇒ Reverse bias connection.

Forward Bias

- * forward bias happens if

$$V_{\text{anode}} > V_{\text{cathode}}$$

OR

$$(V_{\text{anode}} - V_{\text{cathode}} > 0)$$

there is current flow

can be considered as a switch that ON

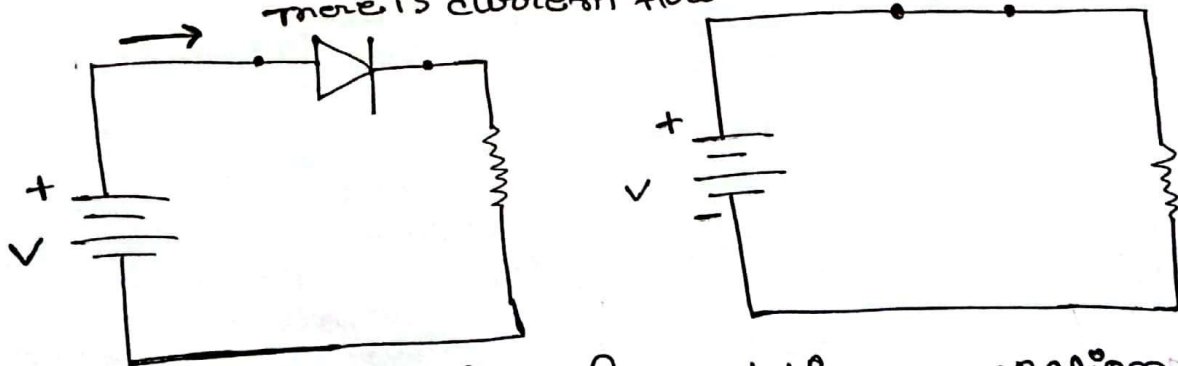


Figure: diode forward bias connection

☐ Reverse Bias

Reverse bias happens if

$$V_{\text{anode}} < V_{\text{cathode}} \text{ On}$$

$$(V_{\text{anode}} - V_{\text{cathode}} < 0)$$

→ no current flow ($I=0$)

can be considered as a switch that is off.

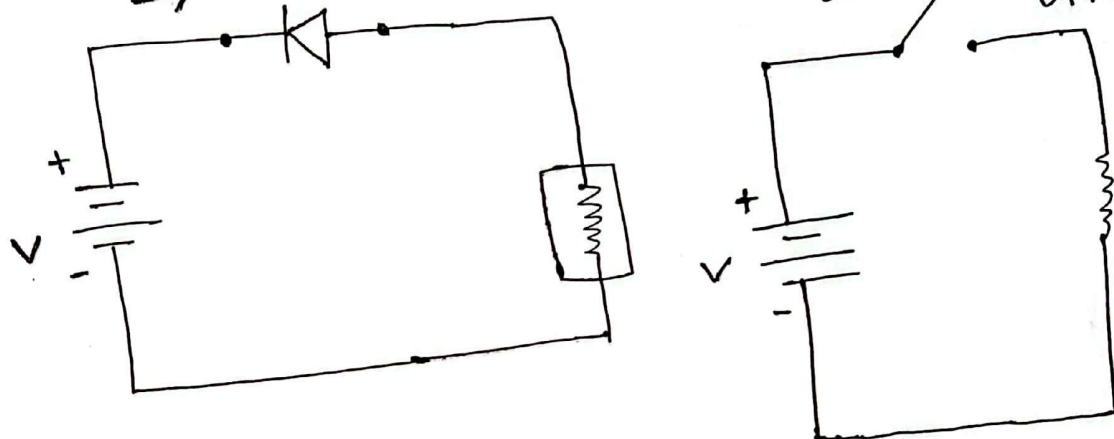


figure: Diode reverse bias connection.

☐ Diode I-v characteristics

Diode I-v characteristics is the relation between the diode current and the voltage difference between anode and cathode of a diode.

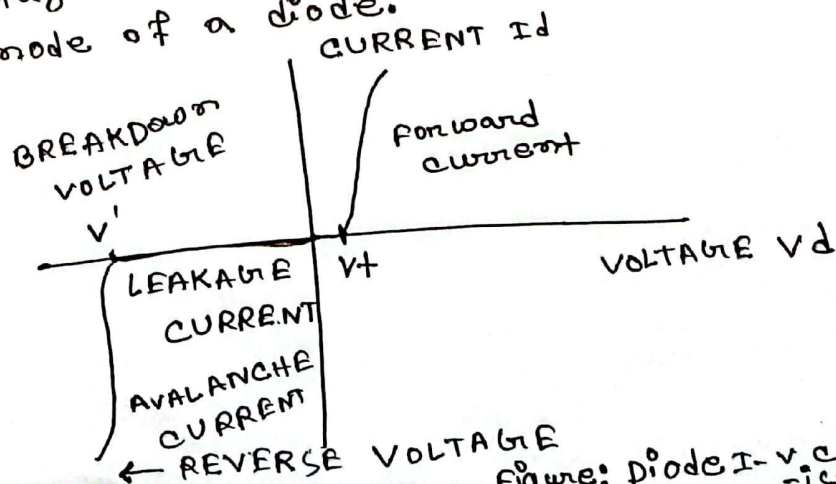


Figure: Diode I-v characteristics.

GREEN UNIVERSITY OF BANGLADESH



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course Code:EEE-204

Course Title:

**ELECTRONIC DEVICE
AND CIRCUITS AND
PULSE TECHNIQUES
LAB**

Experiment

Number:02

Experiment Name :

Study of Clipper

Circuit

Date of Performance:

24.09.2024

Date of Submission

1.10.2024

Submitted to:

Name: Mr.Sakib Abdul Ahad

Lab Lecturer

EEE

Submitted by:

Name: Promod Chandra Das

Dept : CSE

ID: 231002005

Remarks:-

[Handwritten signature and date 01/10/24]

Objectives:

→ To study the diode applications in clipper circuits.

Instruments Required:-

Serial	Name	Rating	Quantity.
1	Function Generator		1
2	Oscilloscope		1
3	DC power supply		1
4	Breadboard, Diodes and Resistors		1

Circuit Diagram:-

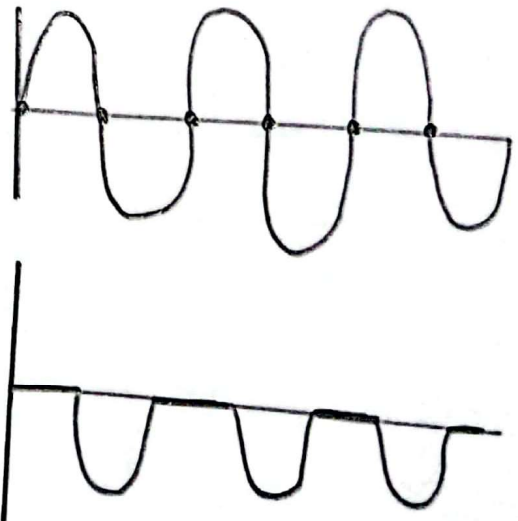
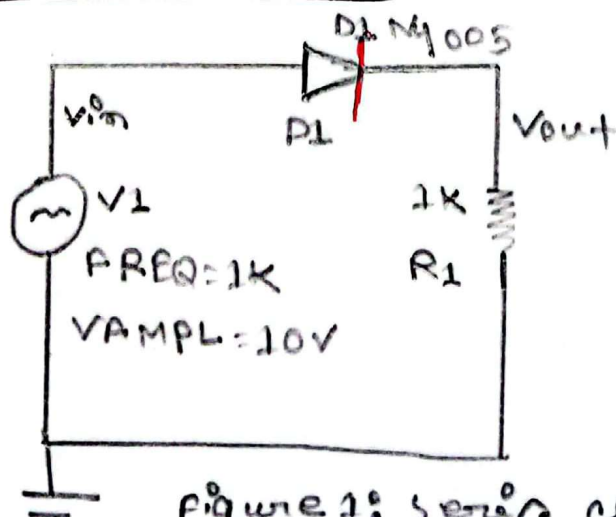


Figure 1: Series clipper (positive)

Theory:

Diode clipper circuits are essential in shaping electrical signals by limiting the voltage levels within a specific range. They work by utilizing the diode's unidirectional conductivity. When the input signal exceeds a predetermined threshold, the diode becomes forward-biased and allows current to flow, clipping off the excess voltage. This process prevents signal distortion and protects downstream components from voltage spikes. Common applications include signal processing in audio systems, over-voltage protection in power supplies, and waveform shaping in communication circuits. By controlling the output waveform, diode clippers enhance system performance, making them vital in various electronic applications.

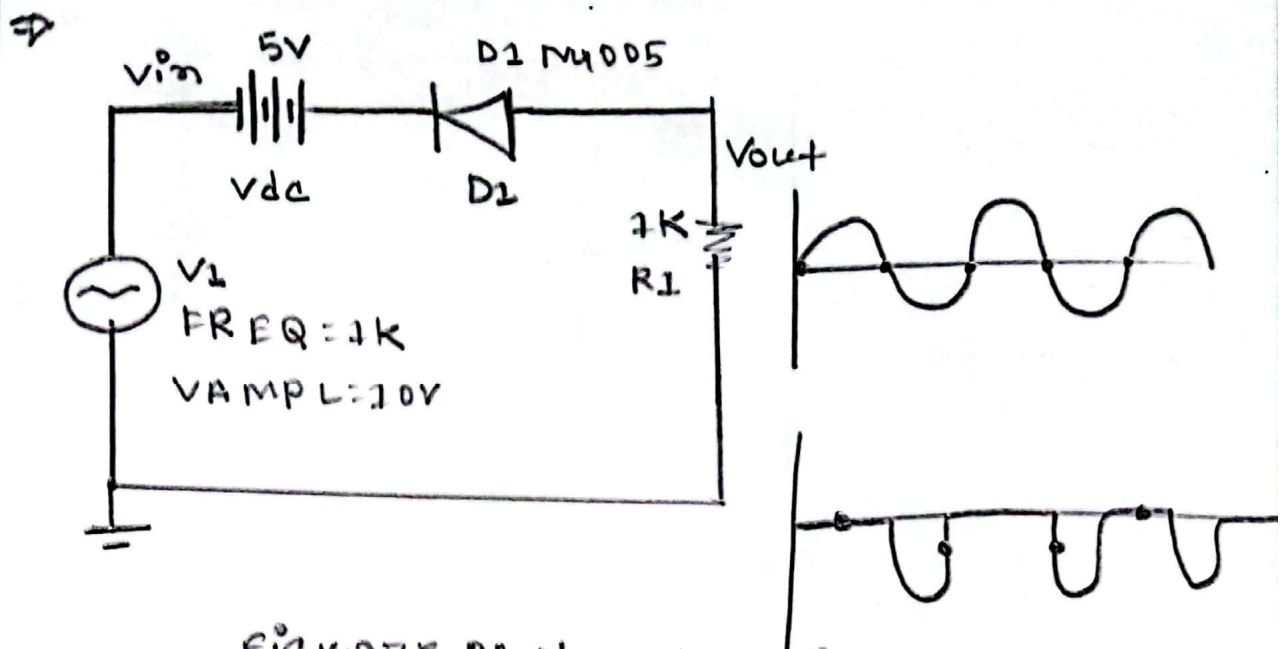


FIGURE 2: Biased series clipper (positive)

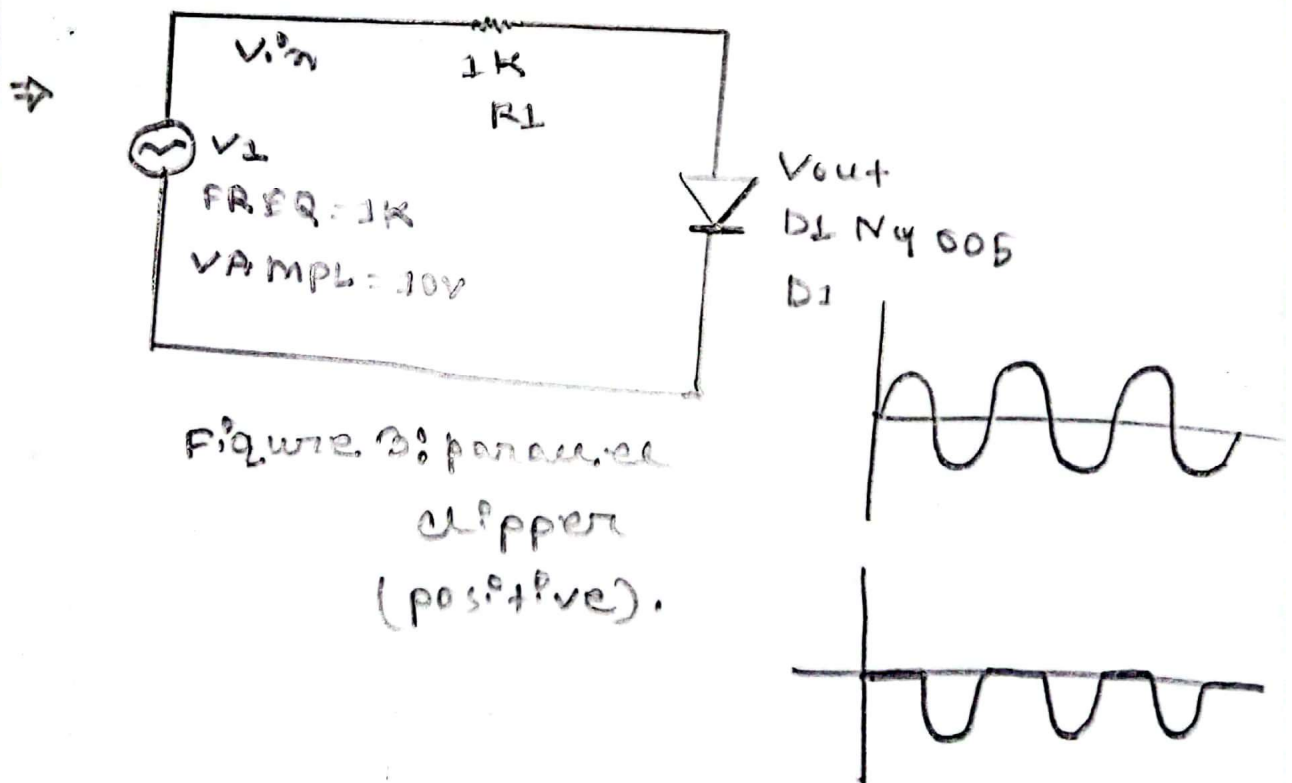


FIGURE 3: parallel clipper (positive).

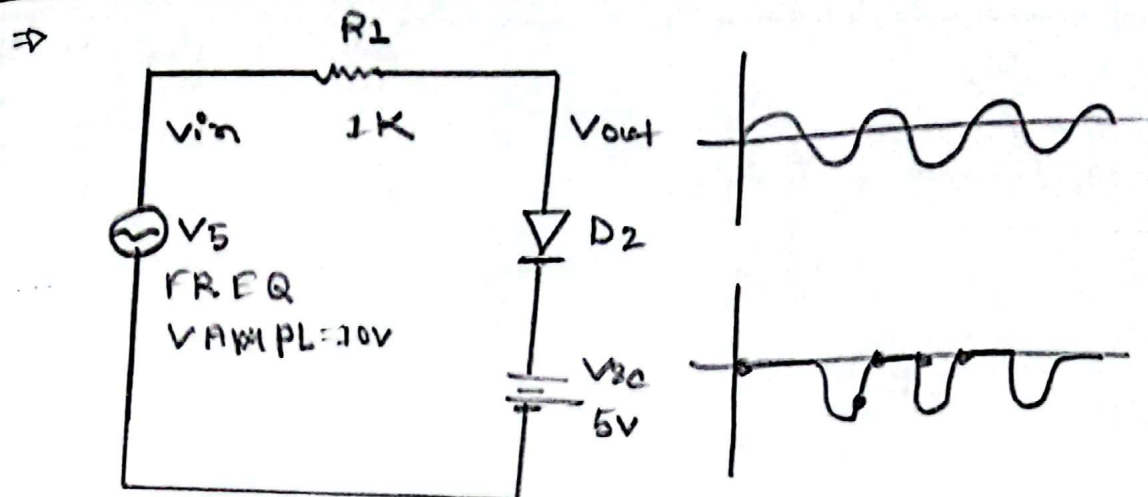


Figure 4: Biased parallel clipper (positive).

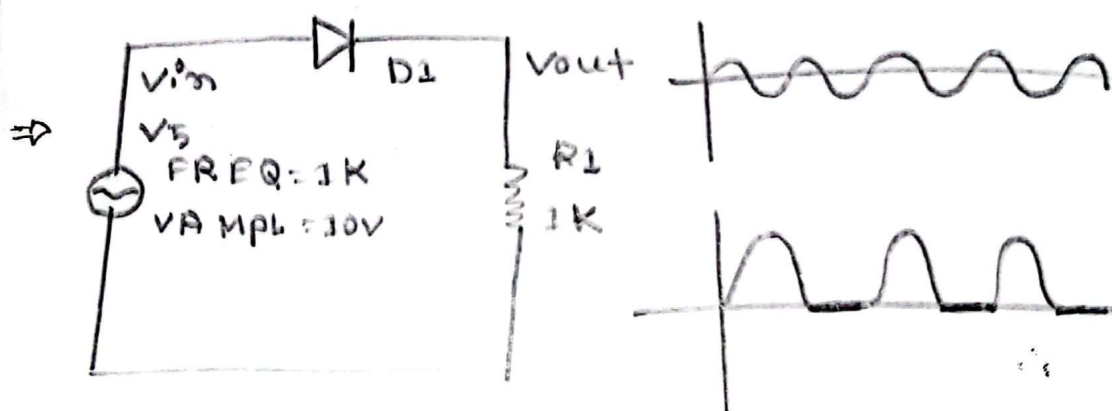


Figure 5: Series clipper (negative).

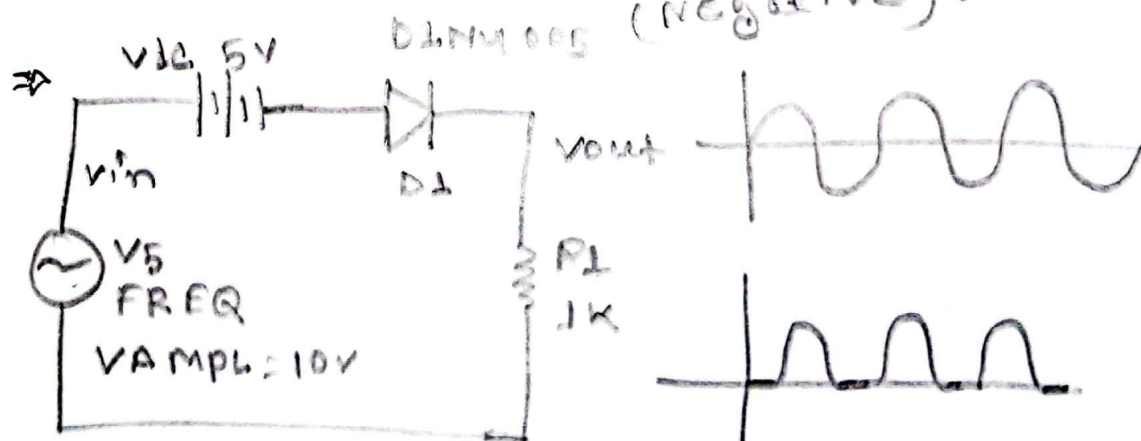


Figure 6: Biased series clipper (negative).

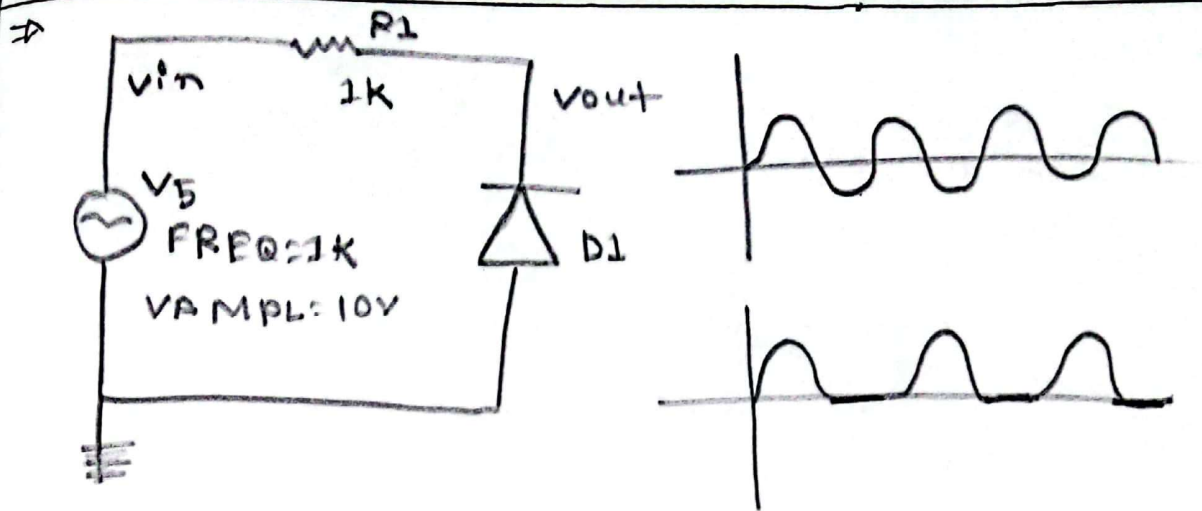


Figure 7: parallel clipper.

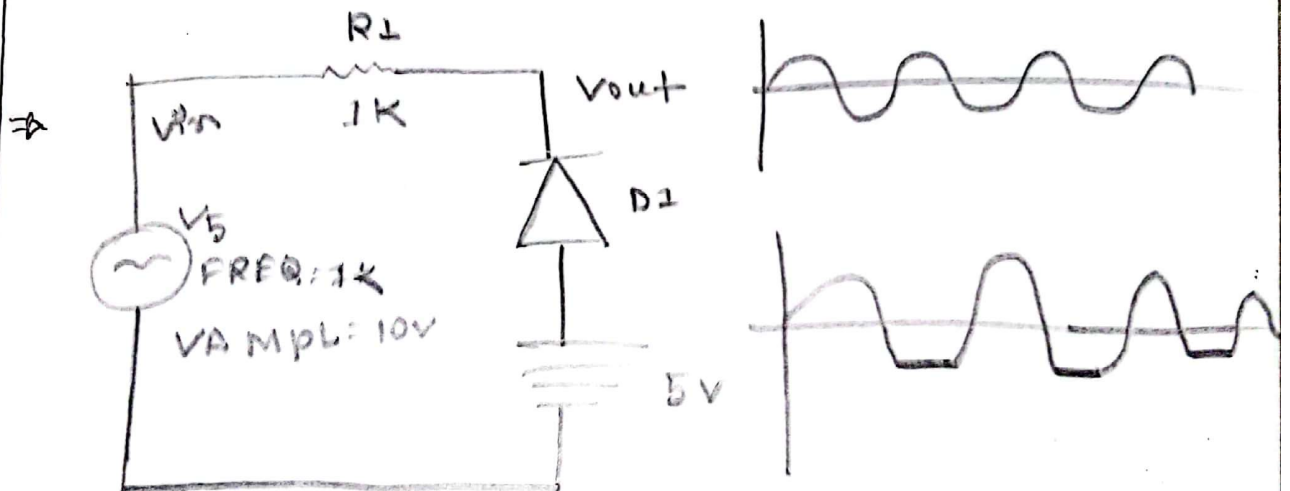


Figure 8: Biased parallel clipper (negative).

Discussion:-

Diode Clipper Circuits are crucial for voltage regulation, waveform shaping and protection in electronic devices. They ensure signal integrity by preventing voltage spikes, thus safeguarding sensitive components from damages.



GREEN UNIVERSITY OF BANGLADESH
Department of Electrical and Electronic Engineering

Course Code: EEE - 202

Course Title: Electronic Devices, Circuits & Pulse Techniques Lab

Experiment Number: 03, Section: 231 D2

Experiment Name: To study the Diode Applications in Clamping circuits

Date of Performance: 3/12/2024

Date of Submission : 10 /12/2024

Submitted to:

Name: Sakib Abdul Ahad

Designation: Lecturer

Department: EEE

Submitted by:

Name : Promod Chandra Das

ID : 231002005

Dept. : CSE

Remarks:

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10.12.24

Objective:

To study the diode applications in clamping circuits.

Required Apparatus:

1. function Generator.
2. oscilloscope.
3. DC power Supply.
4. Breadboard, Diodes, Capacitors and Resistors.

Theory:

A clamping circuit is used to place either the positive or negative peak of signal at a desired level. The dc component is simply added or subtracted to/from the input signal. The clamper is also referred to as an DC restorer and ac signal level shifter.

For a clamping circuit at least three components - a diode, a capacitor and a resistor are required. Sometimes an independent dc supply is also

required to cause an additional shift.
The important points regarding clamping are:

* The shape of the waveform will be the same, but its level is shifted either upward or downward

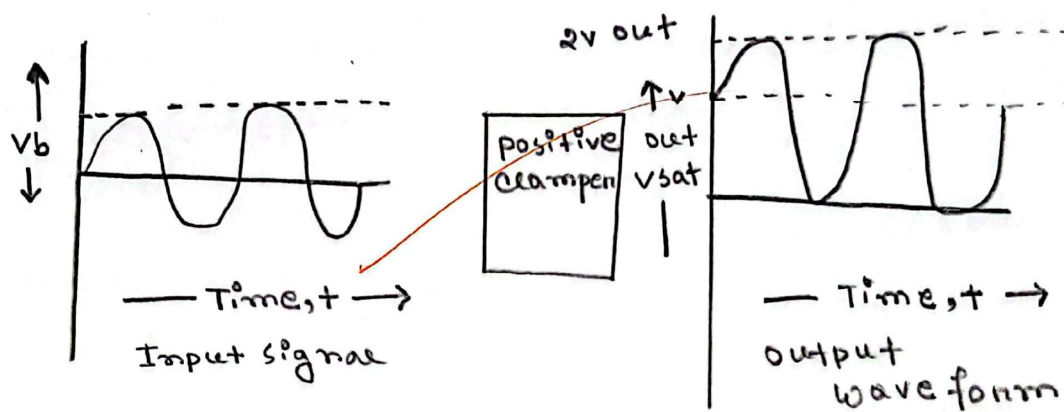
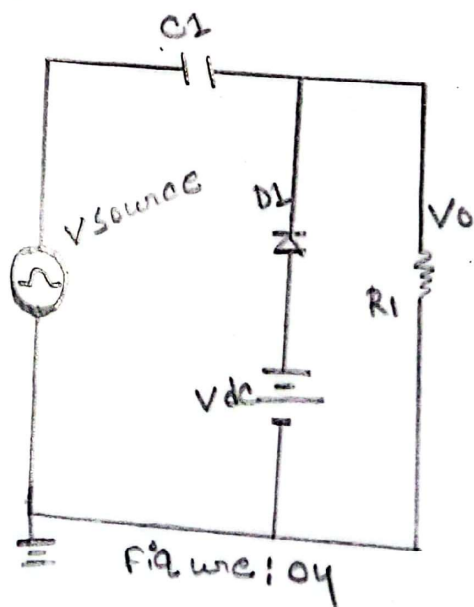
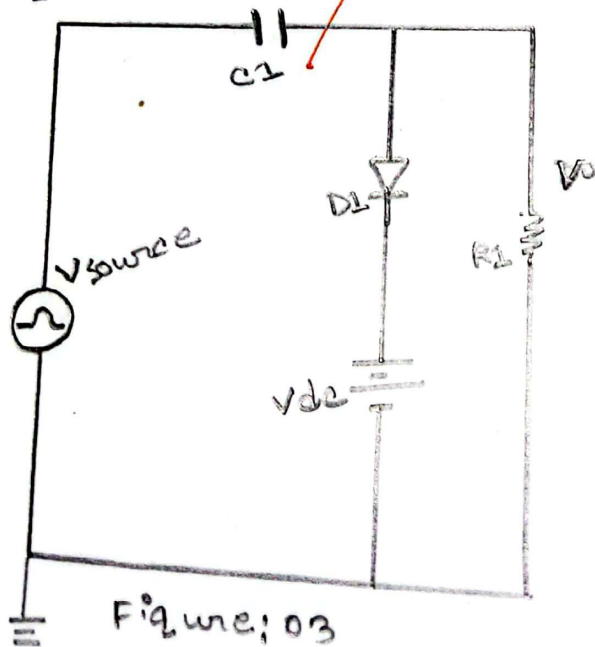
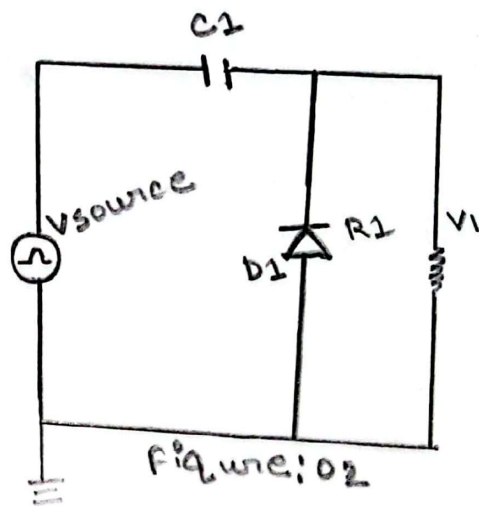
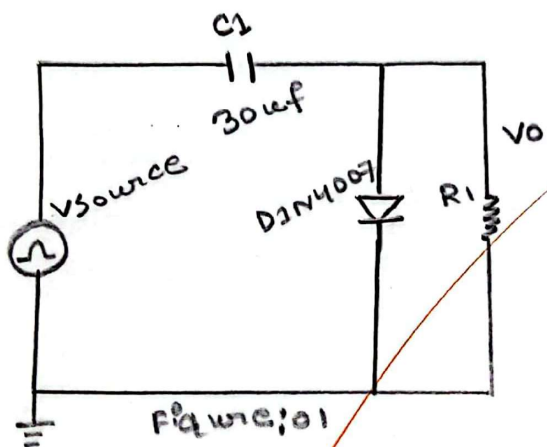


Figure 1: Positive clamper

There will be no change in the peak-to-peak or rms value of the waveform and output waveform will have the same peak-to-peak value that is, $2V_{max}$. This is shown in the figure above, it must also be noted that same reading will be obtained in the ac voltmeter for the input voltage and clamped output voltage.

The voltage across the capacitor does not change significantly during the time interval the diode is non-conducting. In a good clamper circuit, the circuit time constant $\tau = RC$ should be at least ten times the time period of the input signal voltage.

Circuit Diagrams:



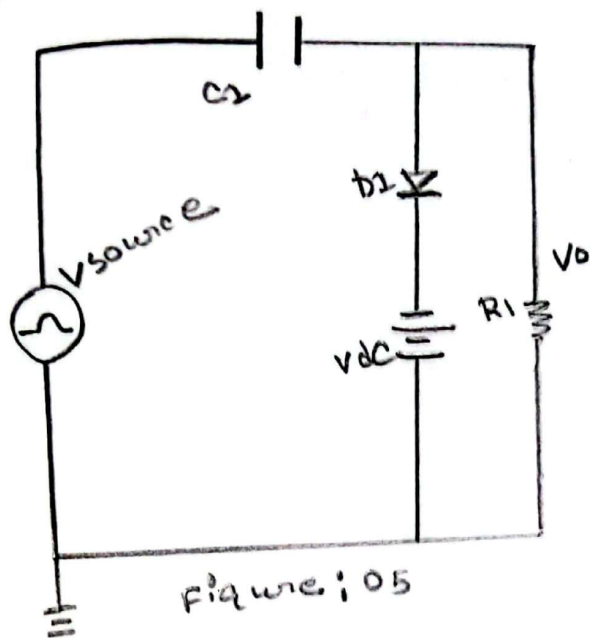


figure: 05

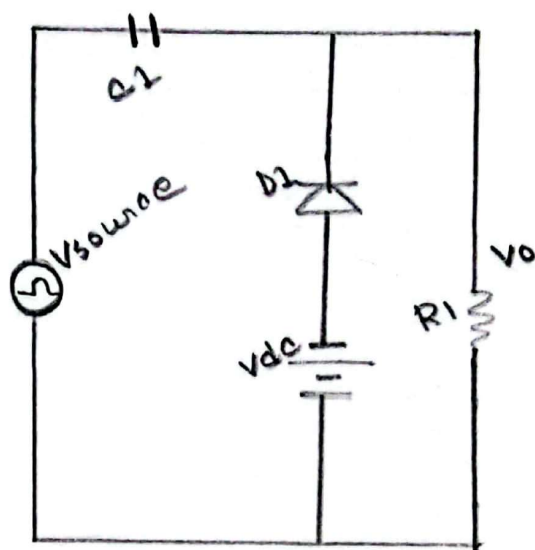


figure: 06

Discussion:

A clamping circuit is type of nonlinear circuit that shifts the voltage level of a signal without changing its shape. It typically uses diodes, capacitors and resistors to clamp the waveform to specific voltage level. clamping circuits are used in signal processing to prevent signal distortion or to reference signals to a desired level.



GREEN UNIVERSITY OF BANGLADESH
Department of Electrical and Electronic Engineering

Course Code: EEE - 202

Course Title: Electronic Devices, Circuits & Pulse Techniques Lab

Experiment Number: 04, Section: 231 D2

Experiment Name: Study of Inverting Amplifier

Date of Performance: 3/12/2024

Date of Submission : 10 /12/2024

Submitted to:

Name: Sakib Abdul Ahad

Designation: Lecturer

Department: EEE

Submitted by:

Name : Promod Chandra Das

ID : 231002005

Dept. : CSE

Remarks:

10.13.24

Objective:-

To design and setup an inverting amplifier circuit with op AMP 741C for a gain of 10 plot the waveforms, observe the phase reversal measure the gain.

Required Apparatus:-

1. Dual power supply
2. Function generator
3. Oscilloscope
4. Breadboard
5. IC 741C
6. Resistors
7. Probs and connecting wires

Theory:

It is a closed loop mode application of opamp and employs negative feedback. The R_f and R_i are the feedback and input resistance of the circuit respectively. The input terminals of the opamp draws no current because of the large differential input impedance. The potential

different across the input terminals of an opamp is zero because of the large open loop gain. Due to these two conditions, the inverting terminal is at virtual ground potential. So the current flowing through R_i and R_f are the same.

$$I_i = I_f$$

that is

$$\frac{V_{in}}{R_{in}} = \frac{V_o}{R_f}$$

therefore

$$\frac{V_o}{V_{in}} = A_v = - \frac{R_f}{R_i}$$

Here the -ve sign indicates that the output will be an amplifier wave with 180° phase shift (inverted output). By varying the R_f or R_i , the gain of the amplifier can be varied to any desired value.

Circuit Diagram:

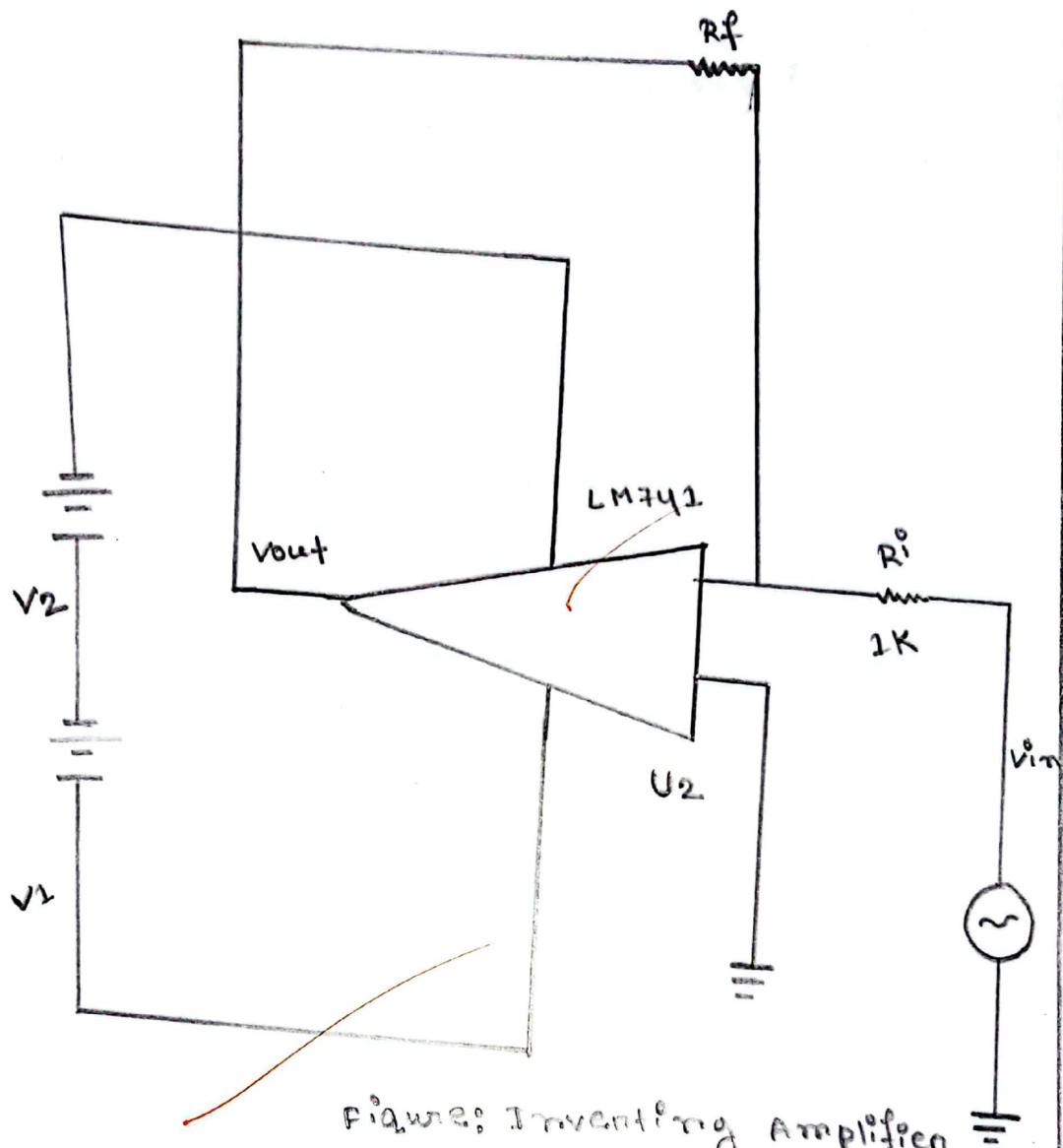


Figure: Inverting Amplifier

Discussion:

An inverting amplifier is a basic op-amp configuration where the input signal is applied to the inverting terminal, producing an output that is inverted and amplified. The gain is determined by the ratio of feedback resistor to input resistor. It's widely used in signal processing, audio applications, and analog computations.



GREEN UNIVERSITY OF BANGLADESH
Department of Electrical and Electronic Engineering

Course Code: EEE - 202

Course Title: Electronic Devices, Circuits & Pulse Techniques Lab

Experiment Number: 05, Section: 231 D2

Experiment Name: Study of Non_ Inverting Amplifier

Date of Performance: 3/12/2024

Date of Submission : 10 /12/2024

Submitted to:

Name: Sakib Abdul Ahad

Designation: Lecturer

Department: EEE

Submitted by:

Name : Promod Chandra Das

ID : 231002005

Dept. : CSE

Remarks:

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10.12.24

☐ Title: Study of non-Inverting Amplifier

☐ Objectives:

To design and setup a non-inverting amplifier circuit with OPAMP IC 741C for a gain of 11 plot the waveform, observe the phase reversal, measure the gain.

☐ Required Apparatus:

1. Dual power supply.
2. Function generator.
3. Oscilloscope
4. Breadboard
5. IC 741C
6. Resistors
7. Probes and connecting wires

☐ Theory:

It is a linear closed loop mode application of op-amp and employs negative feedback. The R_f and R_i are the feedback and input resistance of the circuit respectively. There will be no phase difference between the output and input. Hence it is called non-inverting amplifier.

$$A-V = \frac{V_o}{V_{in}} = 1 + \frac{R_f}{R_i}$$

Here the +ve sign indicates that the output will be amplified wave in phase with the input. By varying the R_f or R_i , the gain of the amplifier can be varied to any desired value.

Circuit diagram:

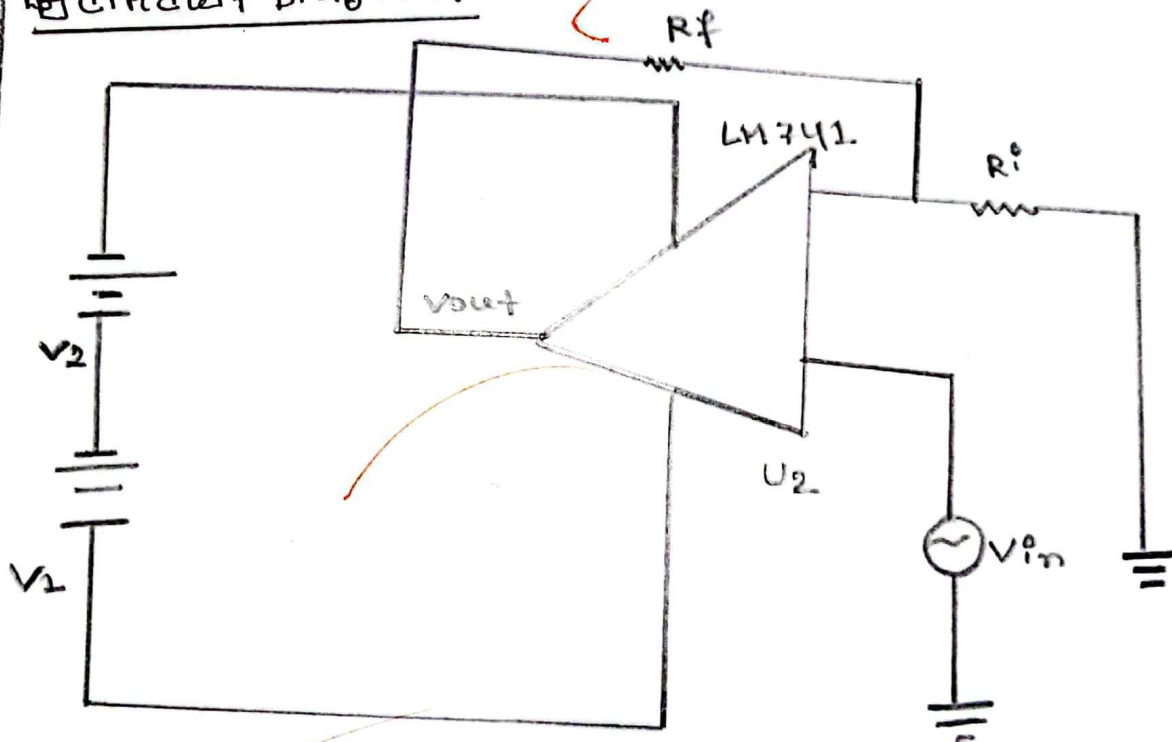


Figure 1: Non-Inverting Amplifier.

Discussion:

An non-inverting amplifier is a fundamental operational amplifier (op-amp) configuration where the input signal is applied to the non-inverting terminal, resulting in an output that is in-phase with the input. The gain of the common amplifier is determined by the ratio of two resistors typically placed in the feedback loop and connected to the inverting terminal. This configuration offers high input impedance, making it ideal for applications where signal integrity must be preserved. Non-Inverting amplifiers are commonly used in audio, instrumentation and control systems where signal amplification without phase inversion is necessary processing and output.